



# Outline of the lecture

## The history of Informatics & HCI

Our roots and evolution

[NAME\_1]

[NAME\_2]

[NAME\_3]

[NAME\_4]

## Action Research & [NAME\_5]

[NAME\_5] (place) studies & ethnography | [NAME\_6]

Praxeology & Appropriation | [NAME\_7]

## Structuration & Infrastructuring

A holistic look at Human-System-Interaction

## HCI in [NAME\_8]

[NAME\_8] | Prototyping | Evaluation

[NAME\_9]

Inform the design of systems that are intuitive, efficient, and aligned with human [NAME\_12]

[NAME\_2] **Model:** Equates the human mind to a computer, where information is processed in stages, including **perception, encoding, storage, and retrieval** - breaks down how users take in information from an interface, interpret it, and decide on actions

[NAME\_12]: Internal representations of **how designers believes a users mind and system works** - built from prior experiences and interactions with similar systems

[NAME\_13]: Cognition not only as an individual process but as something that is distributed across people, artifacts, and the environment - **shared understanding of system accross many cognitions**

**Embodied cognition and affordance:** Thinking and understanding are not just brain-based but also rely on our body's actions, movements, and sensory experiences - Our understanding of objects is rooted in our bodily interactions, and affordances are cues that leverage this understanding, guiding our actions in ways that feel natural and intuitive

[NAME\_9]

aligned with human [NAME\_12]

ient, and

**Basis:** Examination of the physiological and psychological aspects of awareness

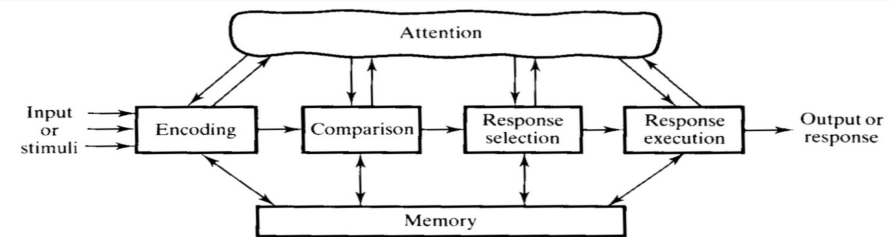
**Aim:** “Good” software supports people in their perception and manipulation of relevant aspects of the environment

**Differentiation:** “How do we perceive?” vs. “How do we understand?”

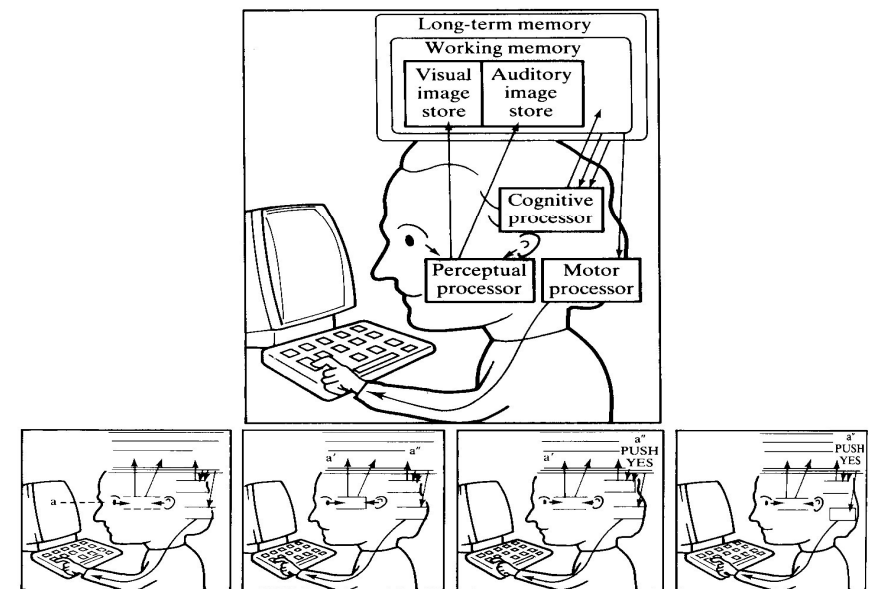
? cognitive scientific approaches

Cognitive considerations are the foundation of software ergonomics / usability engineering

More examples: Attention-Memory model, Human-processor model



**Figure 3.2** Extended stages of the information processing model (adapted from Barber, 1988).



**Figure 3.4** The human processor model (adapted from Card *et al.*, 1983).

# Cognitive: [NAME\_46]

Goal-Directed Design ([NAME\_17])

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Focuses on understanding users' goals and the cognitive steps they take to achieve them. By identifying user goals, tasks are broken down into logical steps that align with how users think.

## What is functional?

- Specific to activity, task or goals
- Practical rather than attractive

## Aim:

- Models of human activities and tasks
- Improve execution
- Reduce error and frustration

## Good for

- Highly structured systems
- Fixed procedures

## Problem is

- Study of task execution may not be enough in some situations
- Not suitable to investigate **wicked problems**

# Methods: [NAME\_10]

Advanced: [NAME\_19] [NAME\_10] [NAME\_20]

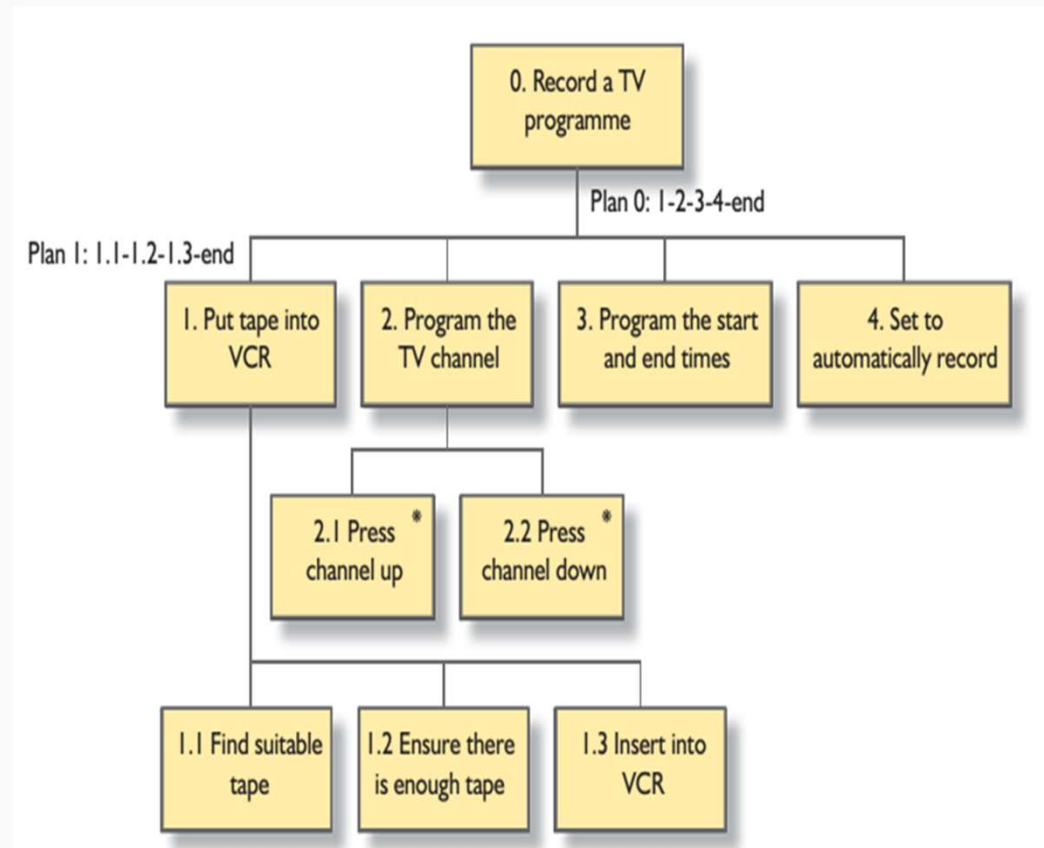
[NAME\_20] Norman (1998) said, users are notoriously bad at clearly articulating goals: e.g., "I want to be a good mom" – where do you even begin? Each subtask should be specified in terms of objectives. Put together, these objectives should cover the whole area of interest—i.e., help a user achieve a goal in full.

**Goals:** State to be achieved

**Tasks:** Goal + [NAME\_25]

[NAME\_25]: Simple performance steps

Once you have laid all the steps out, you will then be in a position to see where additional user support is required (for example, you might wish to automate some actions that the user currently undertakes), or eliminate unnecessary steps, in order to minimize the number of actions that a user has to undertake, unassisted.



# Methods: [NAME\_10]

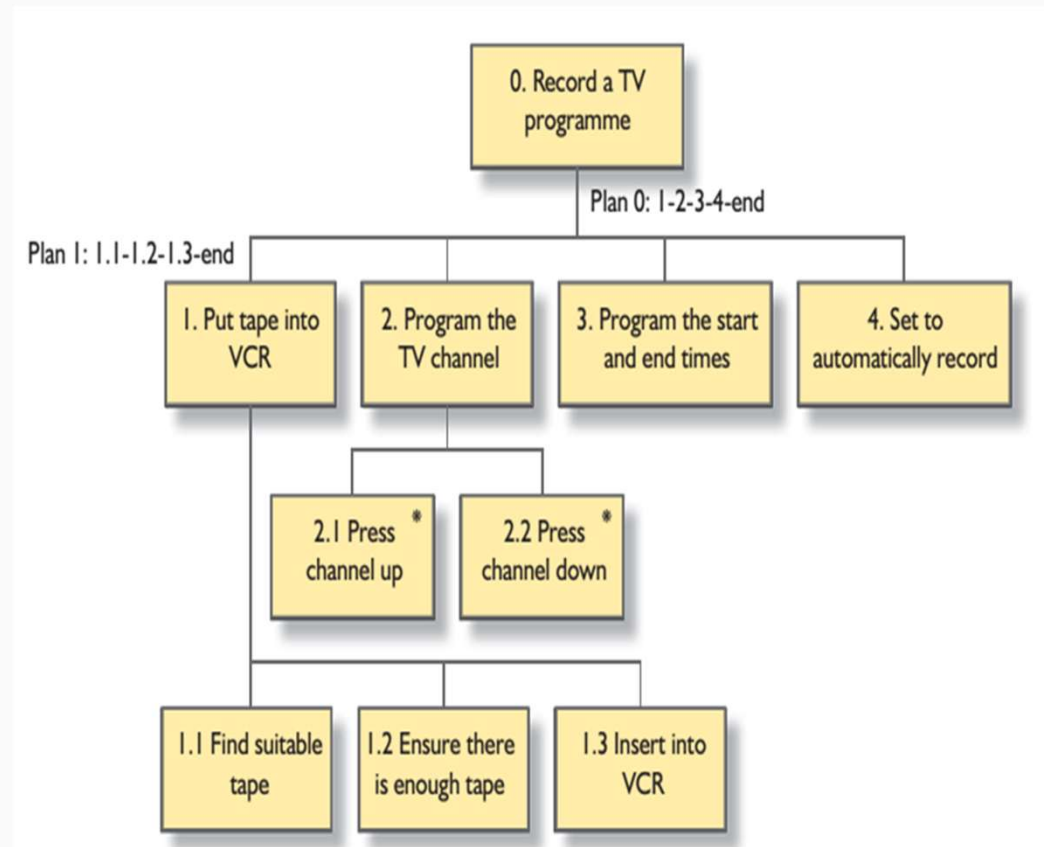
Advanced: [NAME\_20] [NAME\_21] [NAME\_24]

[NAME\_24]: State to be achieved

**Tasks:** Goal + [NAME\_25]

[NAME\_25]: Simple performance steps

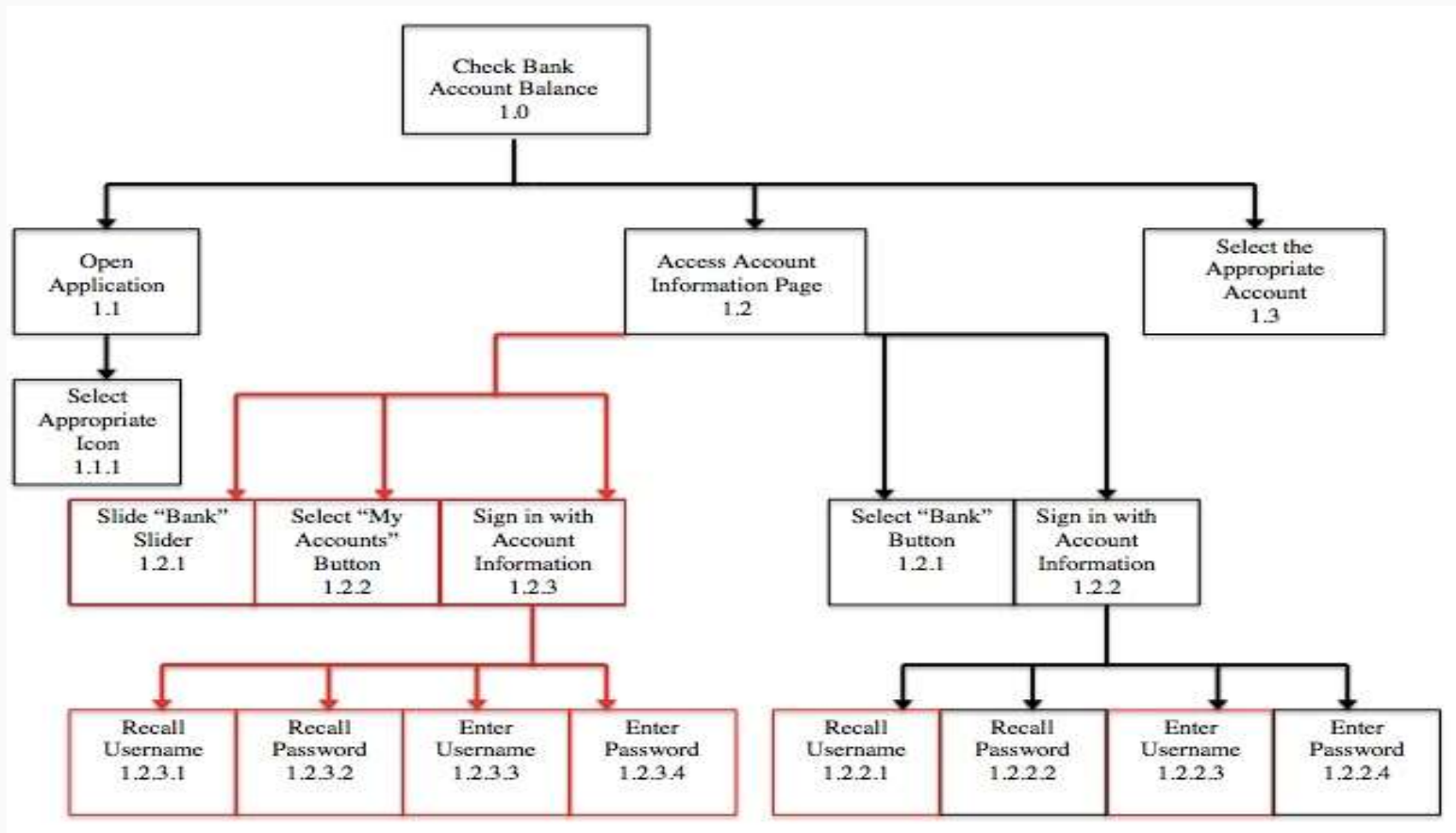
- It is read from top to bottom.
- The rectangles depict the various tasks with the top one representing the overall task.
- From this point, the tree branches out into higher-order subtasks, which, in turn, branch out into sub-subtasks, becoming increasingly granular until the level of “interactions” is reached (e.g., “Insert into VCR”).
- So-called Plans specify the sequence in which tasks on the same hierarchical level are carried out.
- Tasks may be executed in order based on their sequence and numbering, in any order, or only if certain conditions are met.



(Benyon 2010)

# Methods: [NAME\_20]

Advanced: [NAME\_21] [NAME\_20]

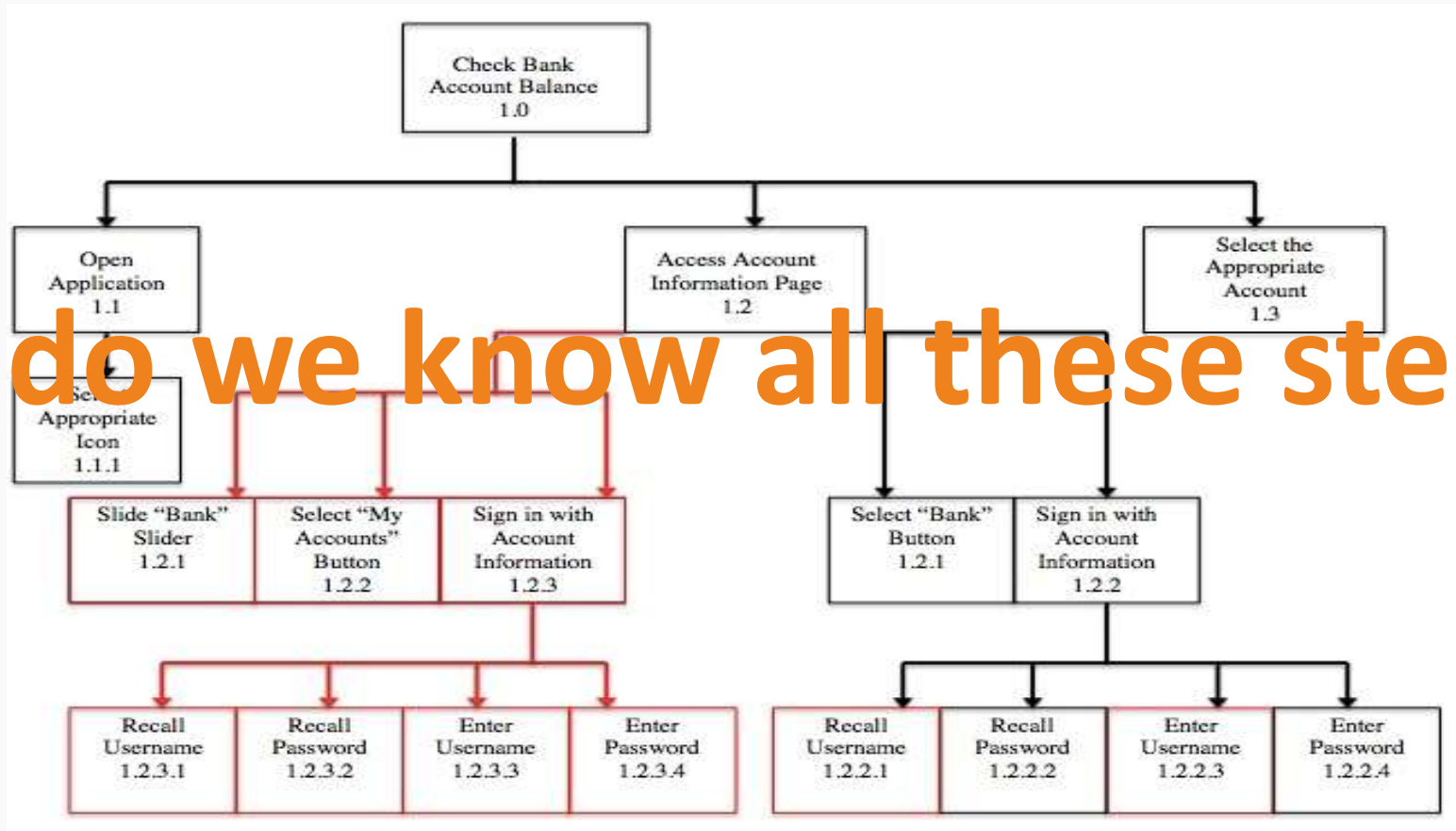


(<https://hfacmethods.wordpress.com/hierarchical-task-analysis/>)

# Methods: [NAME\_20]

Advanced: [NAME\_21] [NAME\_20]

How do we know all these steps?



(<https://hfacmethods.wordpress.com/hierarchical-task-analysis/>)



# Methods: [NAME\_20]

Advanced: [NAME\_21] [NAME\_22] [NAME\_23]

## Goal: Send an Email

- 1.1 [NAME\_33]
  - 1.1.1 [NAME\_34] Client
  - 1.1.2 [NAME\_35] Message
  - 1.1.3 [NAME\_36]
  - 1.1.4 [NAME\_37] Content
  - 1.1.5 [NAME\_38] (if necessary)
  - 1.1.6 [NAME\_39]
- 1.2 [NAME\_40]
  - 1.2.1 [NAME\_41]
  - 1.2.2 [NAME\_42] Button
  - 1.2.3 Wait for Confirmation
- 1.3 [NAME\_43] [NAME\_44]
  - 1.3.1 Save a Copy in [NAME\_45]
  - 1.3.2 [NAME\_46] Client

# Methods: [NAME\_20]

Advanced: [NAME\_21] [NAME\_22] [NAME\_23]

[NAME\_49]

- Hard to accommodate tasks completed by more than one individual, except in very simple cases.
- Representation of a [NAME\_30] is complex, even when a simple task is studied.
- Complex tasks (e.g. air traffic control) become very unwieldy very fast, but can be done, but their interpretation is often limited to those that conducted the analysis.

| [NAME_31] [NAME_32]                                                                                                                                                                                                                                                                                                                                                                                                            | [NAME_50] [NAME_51] [NAME_52]                                                                                                                                                                                                                                                   | Logical [NAME_33]                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Involves decomposing tasks into subtasks and representing the order and structure through structure diagrams.</li><li>• Performing HTA is a time-consuming process with many choices.</li><li>• It is fussy, and takes effort when you could be having fun coding or chatting to users.</li><li>• In essence it is a <b>representational device</b> rather than a technique.</li></ul> | <ul style="list-style-type: none"><li>• [NAME_50] [NAME_34] focuses on what the user knows and frequently is concerned with the quality of the <b>mapping</b> between a representation of the user's knowledge and that required by the system.</li><li>• GOMS family</li></ul> | <ul style="list-style-type: none"><li>• Sometimes one wants to analyse what a system as a whole must do, not just what the user must do, or just what the system must do (though it is also called <b>conceptual design</b>).</li><li>• This can be especially useful if one is designing a new way to do a familiar task, since the logical description should apply equally to the before and after systems.</li></ul> |

# Methods: GOMS

One of the first “Usability” methods (not so common today)

Quantification of the use efficiency of systems over time which require interaction. Foundations: systematically developed tables (“A mouse click takes X seconds, keyboard entries take N seconds”,...). Problem: **Decoupling of qualitative and quantitative aspects, separation of functionality and interface.**



Goals



Operations

Concrete operations (Keystrokes etc.)



[NAME\_53]

[NAME\_53] sequence of operations



[NAME\_54]

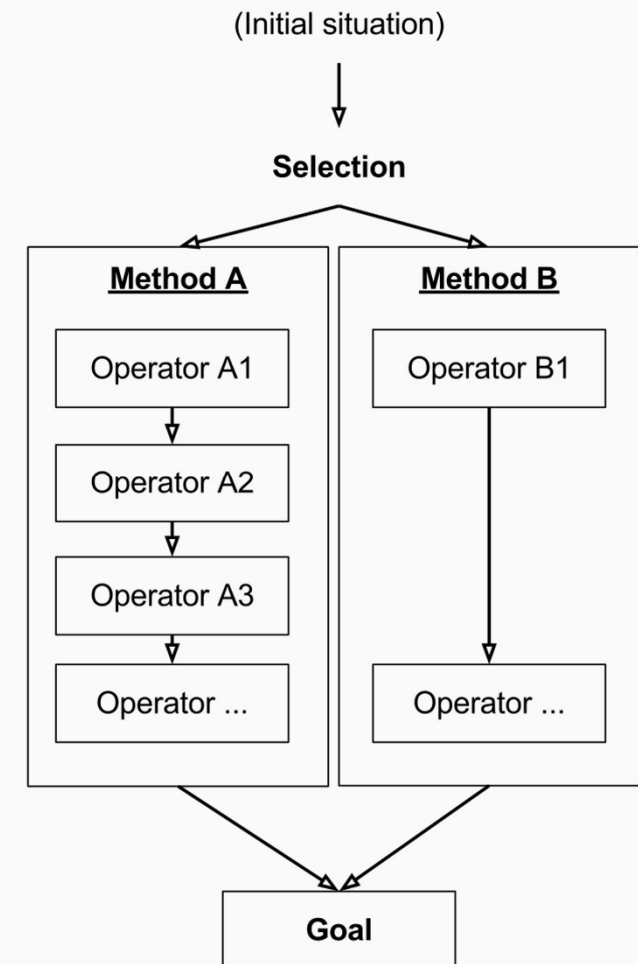
Choices

Selection

Method

Operator

Goal



# Methods: GOMS

One of the first “Usability” methods (not so common today)

Quantification of the use efficiency of systems over time which require interaction. Foundations: systematically developed tables (“A mouse click takes X seconds, keyboard entries take N seconds”,...). Problem: **Decoupling of qualitative and quantitative aspects, separation of functionality and interface.**

- G** Goals
- O** Operations  
Concrete operations (Keystrokes etc.)
- M** [NAME\_53]  
[NAME\_53] sequence of operations
- S** [NAME\_54]  
Choices

(John & Kieras 1996)

## Moving text with the MENU-METHOD

| Description                                                      | Operator | Duration (sec) |                                                                                                  |
|------------------------------------------------------------------|----------|----------------|--------------------------------------------------------------------------------------------------|
| Mentally prepare by Heuristic Rule 0                             | M        | 1.35           | mark text to be moved                                                                            |
| Move cursor to beginning of phrase<br>(no M by Heuristic Rule 1) | P        | 1.10           |                                                                                                  |
| Click mouse button<br>(no M by Heuristic Rule 0)                 | K        | 0.20           |                                                                                                  |
| Move cursor to end of phrase<br>(no M by Heuristic Rule 1)       | P        | 1.10           |                                                                                                  |
| Shift-click mouse button<br>(one average typing K)               | K        | 0.28           | TWO commands needed to complete a move. Should we consider a MOVE command instead?               |
| (one mouse button click K)                                       | K        | 0.20           |                                                                                                  |
| Mentally prepare by Heuristic Rule 0                             | M        | 1.35           | issue CUT command                                                                                |
| Move cursor to Edit menu<br>(no M by Heuristic Rule 1)           | P        | 1.10           |                                                                                                  |
| Press mouse button                                               | K        | 0.10           | indicate insertion point                                                                         |
| Move cursor to Cut menu item<br>(no M by Heuristic Rule 1)       | P        | 1.10           |                                                                                                  |
| Release mouse button                                             | K        | 0.10           | issue PASTE command                                                                              |
| Mentally prepare by Heuristic Rule 0                             | M        | 1.35           |                                                                                                  |
| Move cursor to insertion point                                   | P        | 1.10           | Issuing commands will be used a LOT! Can we shorten this procedure? Consider keyboard shortcuts. |
| Click mouse button                                               | K        | 0.20           |                                                                                                  |
| Mentally prepare by Heuristic Rule 0                             | M        | 1.35           |                                                                                                  |
| Move cursor to Edit menu<br>(no M by Heuristic Rule 1)           | P        | 1.10           |                                                                                                  |
| Press mouse button                                               | K        | 0.10           |                                                                                                  |
| Move cursor to Paste menu item<br>(no M by Heuristic Rule 1)     | P        | 1.10           |                                                                                                  |
| Release mouse button                                             | K        | 0.10           |                                                                                                  |
| <b>TOTAL PREDICTED TIME</b>                                      |          | <b>14.38</b>   |                                                                                                  |

Fig. 2. A Keystroke-Level Model for moving the text in Figure 1. The notes on the right represent handwritten notes an analyst might add to the KLM to highlight ideas.

# Methods: GOMS

Goal: Send an [NAME\_57]

[NAME\_57]:

K (Keystroke) for keyboard actions.

P (Pointing) for mouse actions.

H (Homework) for mental operations.

## Methods:

M1: [NAME\_58]

K: [NAME\_34] client (Ctrl + E)

K: Create a [NAME\_68] (Ctrl + N)

H: Think about the recipient's email address and type it.

K: Compose the email content.

K: [NAME\_38] (Ctrl + Shift + A if needed)

K: Proofread the message.

M2: [NAME\_59]

K: Review the email.

P: Move the mouse pointer to the "Send" button.

P: Click the "Send" button.

H: Wait for a confirmation message.

M3: [NAME\_60] [NAME\_61]

K: Save a copy in the [NAME\_61] folder (Ctrl + S).

K: Close the email client (Ctrl + W).

[NAME\_62]:

Use the keyboard for text entry and shortcut commands.

Use the mouse for pointing and clicking.

Use mental operations for thinking and decision-making.

[NAME\_63]

[NAME\_64] [NAME\_65]:

- Create a [NAME\_66] [NAME\_67] for completing a purchase on an e-commerce website.
- Start from selecting an item in the shopping cart to completing payment.
- Consider subtasks like selecting a delivery address, choosing a payment method, and confirming the order.

**Booking a [NAME\_68]:**

- Perform HTA on booking a flight through an airline website.
- Decompose tasks from searching for flights to completing payment.
- Identify conditional tasks, such as choosing one-way versus round-trip, selecting seats, and adding baggage.

[NAME\_69] a [NAME\_70] Account:

- Analyze the task of [NAME\_71] a [NAME\_72] account on a web-based platform.
- Identify and break down steps, from filling in personal details to configuring security settings.
- Include optional tasks, such as enabling two-factor authentication.

**Logging into a Website:**

- Develop a GOMS model for logging into a website.
- Define the **goal** (accessing the account), **operators** (entering username, password, clicking “[NAME\_73]”), and possible **methods** (using password autofill, manually entering credentials).
- Suggest [NAME\_74] based on conditions, such as using autofill when available or manually entering credentials if autofill is disabled.

**Sending a Message on a [NAME\_75]:**

- Create a GOMS model for sending a message on a mobile [NAME\_76].
- Define the **goal** (sending a message to a contact), **operators** (opening the app, selecting the contact, typing, pressing “Send”), and **methods** (typing versus voice input).
- Suggest [NAME\_77] based on factors like context (voice input if typing is not feasible, or typing if in a public place).

**Saving a File in a Word Processor:**

- Analyze the task of saving a document in a word processor using GOMS.
- Define the **goal** (save the document), **operators** (clicking “File,” choosing “[NAME\_78],” entering a filename), and **methods** (keyboard shortcuts or menu navigation).
- Identify [NAME\_79] based on user familiarity (experienced users might use shortcuts, while beginners use the menu).

# The HCI [NAME\_72]

[NAME\_72]

Analytic  
Design/Implementation

*Human computer interaction is a difficult endeavor with glorious rewards.*

Designing interactive systems to be

effective  
efficient  
easy to use  
enjoyable to use

*so that **people** and **society** may realize the benefits of computation-based devices*

## How to go beyond platitudes like “know the user”?

# Users and [NAME\_78]

[NAME\_74]

| [NAME_76] [NAME_77]                             | [NAME_75] [NAME_76]                                                                                  | [NAME_74] [NAME_75]                                                                     |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Those who directly interact with the technology | Those who use it via an intermediary; may have an influence about primary users (using/not using it) | [NAME_77] who are affected by the technology use and decide its purchase/implementation |

[NAME\_78]

- Where are users located?
- What are the social, cultural and organisational aspects of the context?
- With what/whom do they interact?
- What role do technologies play in the context?
- How do these aspects impact upon the way that users do things?

stakeholders

motivations, and goals

understand people's

behaviors

An individual or group with an interest in the success of a product/organization in delivering intended results and maintaining the viability of the organization's products and services. **Stakeholders influence** programs, products, and services (Kotonya and Sommerville, 1998)